

# Managing Memory Tiers with CXL in Virtualized Environments

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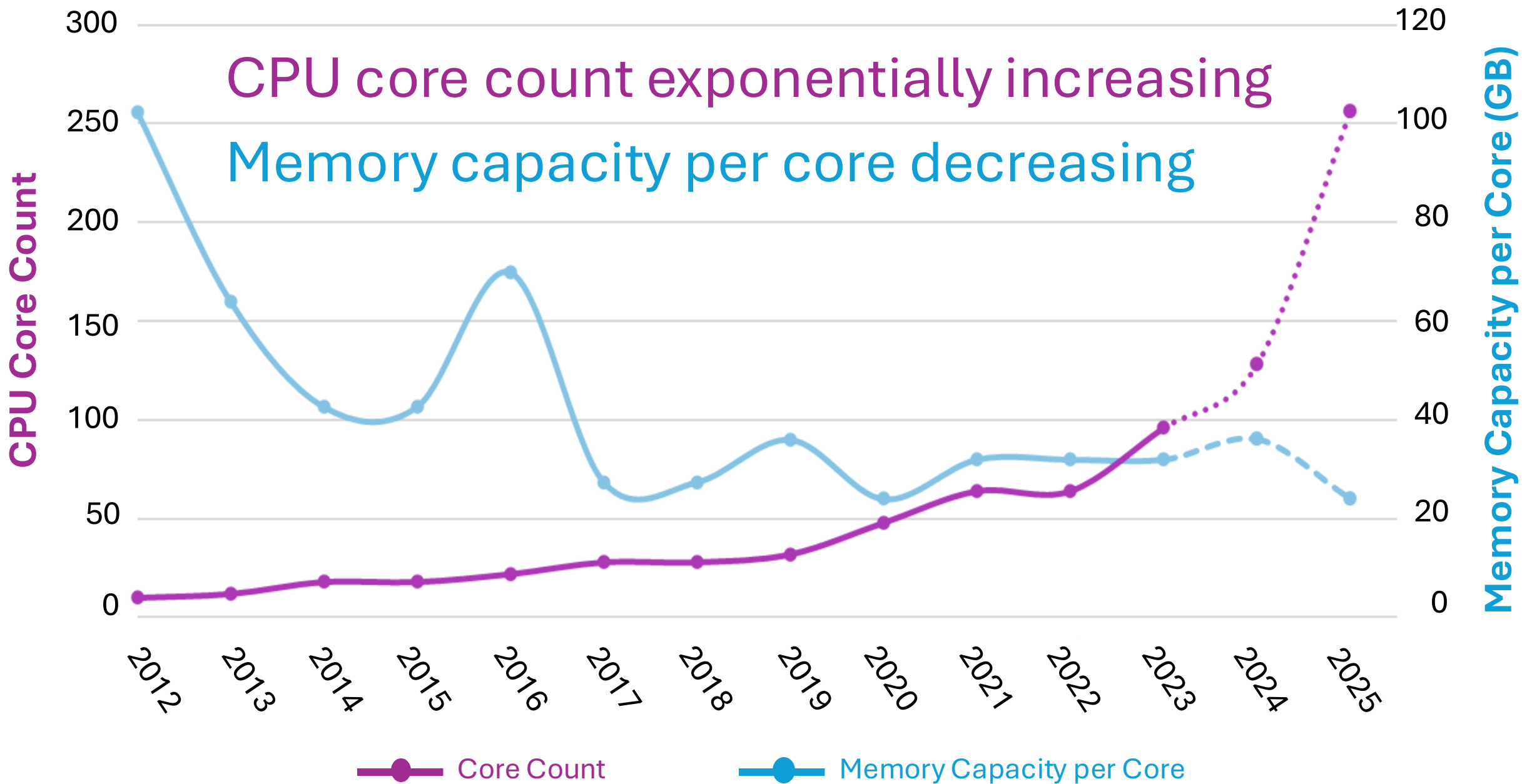
# Executive Summary

## Background:

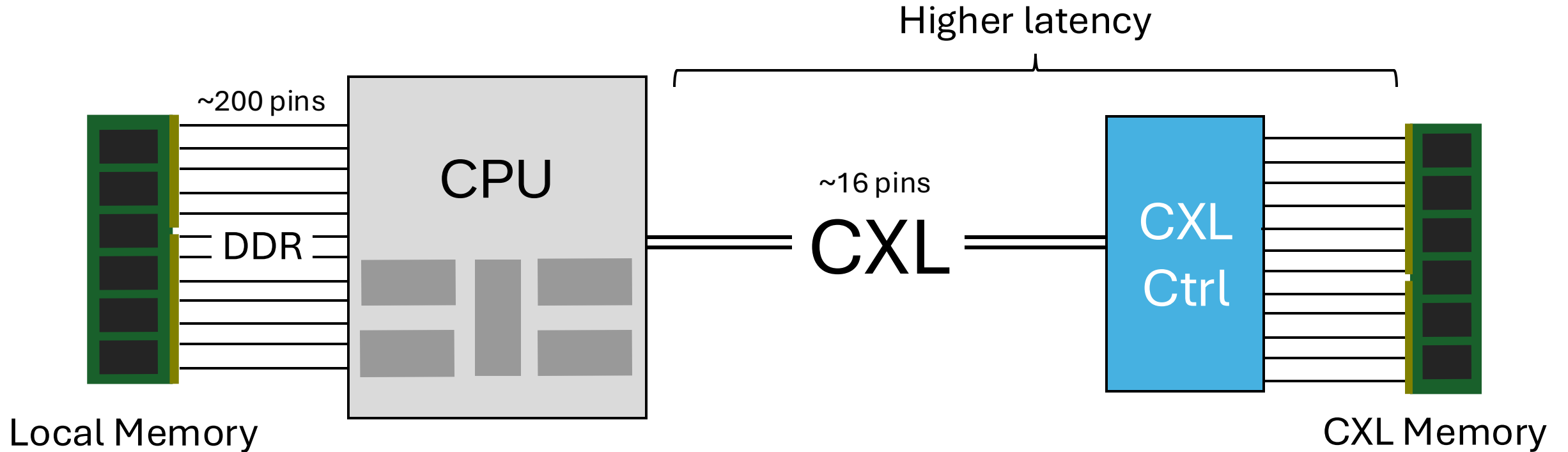
- CPU **core counts scaling faster** than memory capacity
- CXL enables **second-tier memory** to facilitate core scaling
- But CXL adds latency that hurts performance if not mitigated
- Software tiering helps some but **is not well suited for public clouds**

## Contributions:

- Intel Flat Memory Mode: First **hardware-managed** memory tiering for CXL
  - But still has **limitations** that degrade workloads
- Memstrata: Memory allocator for hardware tiering to **mitigate outliers**
- Slowdown reduces to ~5% vs. unattainable one-tier memory



# CXL Enables Memory Capacity Scaling



# Higher CXL Latency Can Degrade Workloads

- CXL latency (220 ns)  $\approx$  2x local memory latency (100 ns)
- CXL slowdowns workloads by up to 62%
- Memory tiering: place data between local and CXL memory

Cloud requirements for CXL include:

- **Minimal slowdown**
- **Low CPU overhead**
- **Huge page friendly**

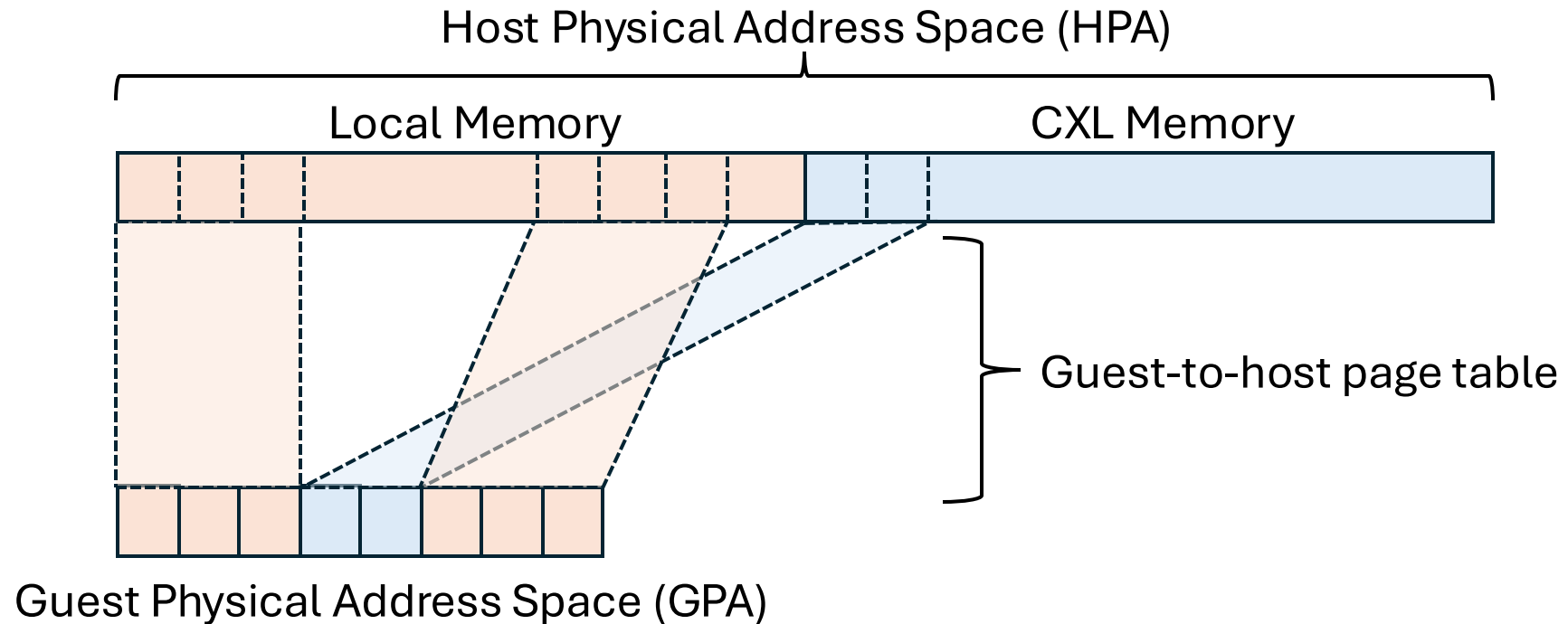
# Combining Software and Hardware for Memory Tiering

|                    | Software Tiering                                          | Hardware Tiering                  | Software + Hardware Tiering                    |
|--------------------|-----------------------------------------------------------|-----------------------------------|------------------------------------------------|
|                    | HeMem (SOSP '21)<br>TPP (ASPLOS '23)<br>MENTIS (SOSP '23) | <b>Intel Flat<br/>Memory Mode</b> | Intel Flat Memory Mode and<br><b>Memstrata</b> |
| Minimal slowdown   | ⚠ High tail slowdown                                      | ⚠ High tail slowdown              | ✅ Minimal slowdown                             |
| Low CPU overhead   | ❌ High overhead                                           | ✅ Low overhead                    | ✅ Low overhead                                 |
| Huge page friendly | ❌ Unfriendly                                              | ✅ Friendly                        | ✅ Friendly                                     |

Introduced in this work

# Prior Work: Software-Managed Memory Tiering

Use **hypervisor/OS** to **identify popular pages** and **decide page placement**



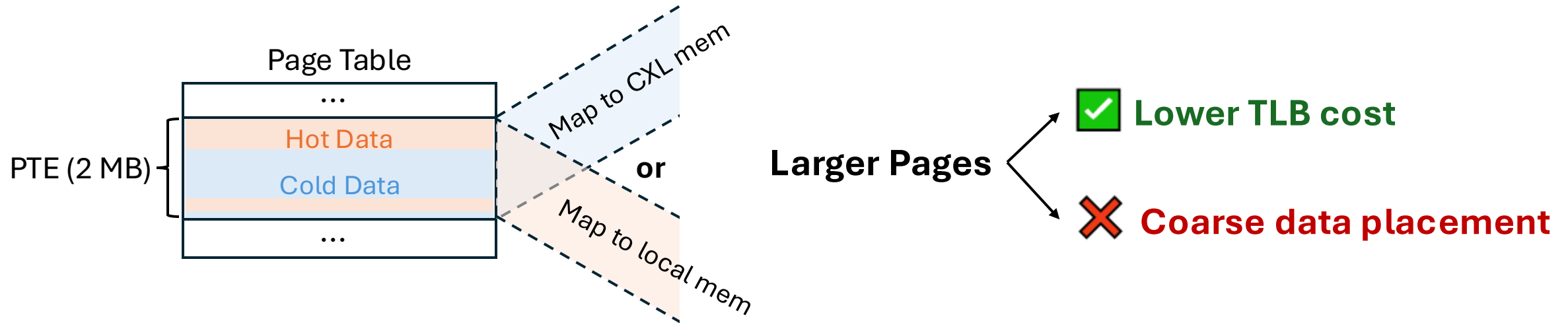
# Software Tiering at Odds With Virtualization

## Issue 1: High CPU overhead

- Instruction sampling (PEBS, IBS) is disabled in clouds
- Frequent page table scans incur excessive CPU overhead

## Issue 2: Huge page penalty<sup>[1]</sup>

- Virtualization uses larger page sizes (2 MB, 1 GB) to reduce TLB cost





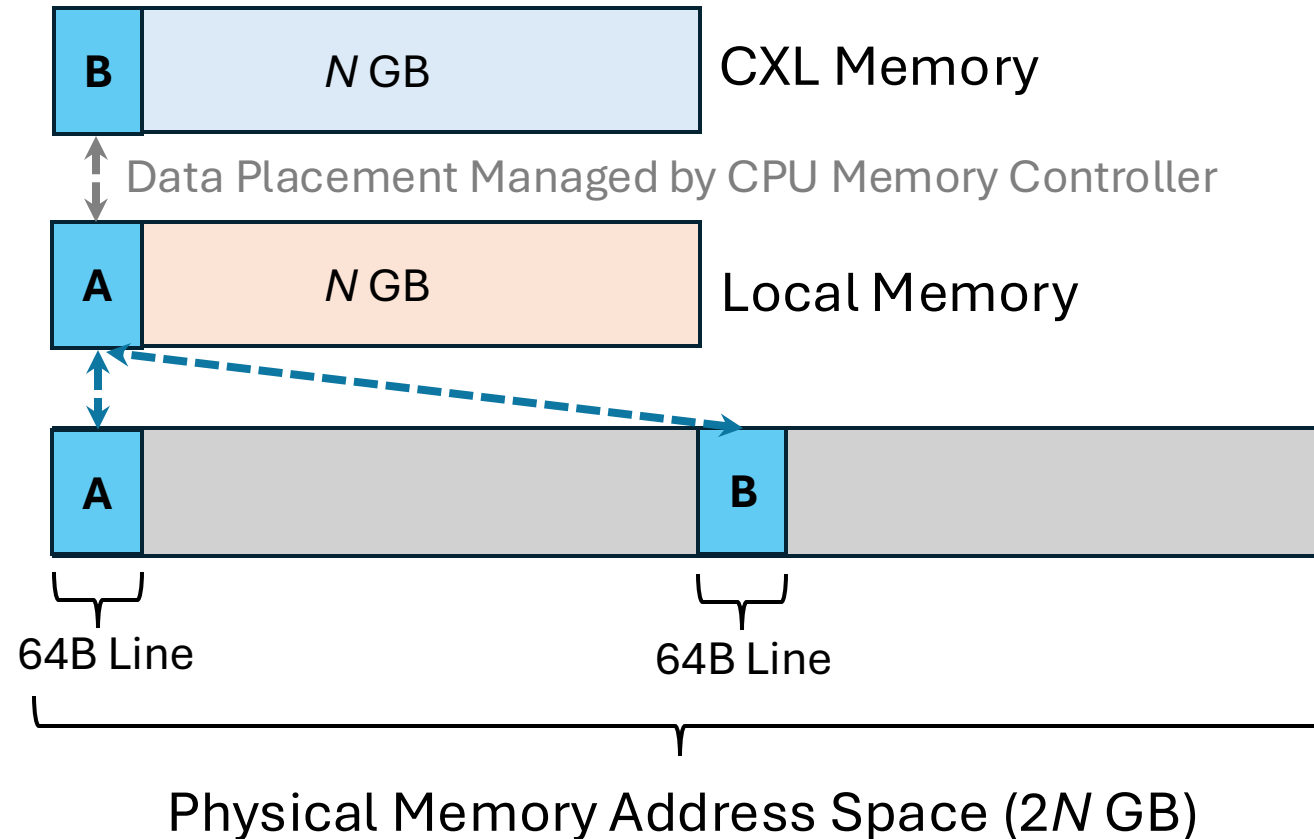
# Introducing Hardware Tiering for CXL

We introduce **Intel Flat Memory Mode**:

- First hardware-managed cacheline-granular memory tiering for CXL
- Data placement managed by the CPU memory controller
  - Zero CPU overhead
  - Huge page friendly
- Available in Intel Xeon 6 Processor

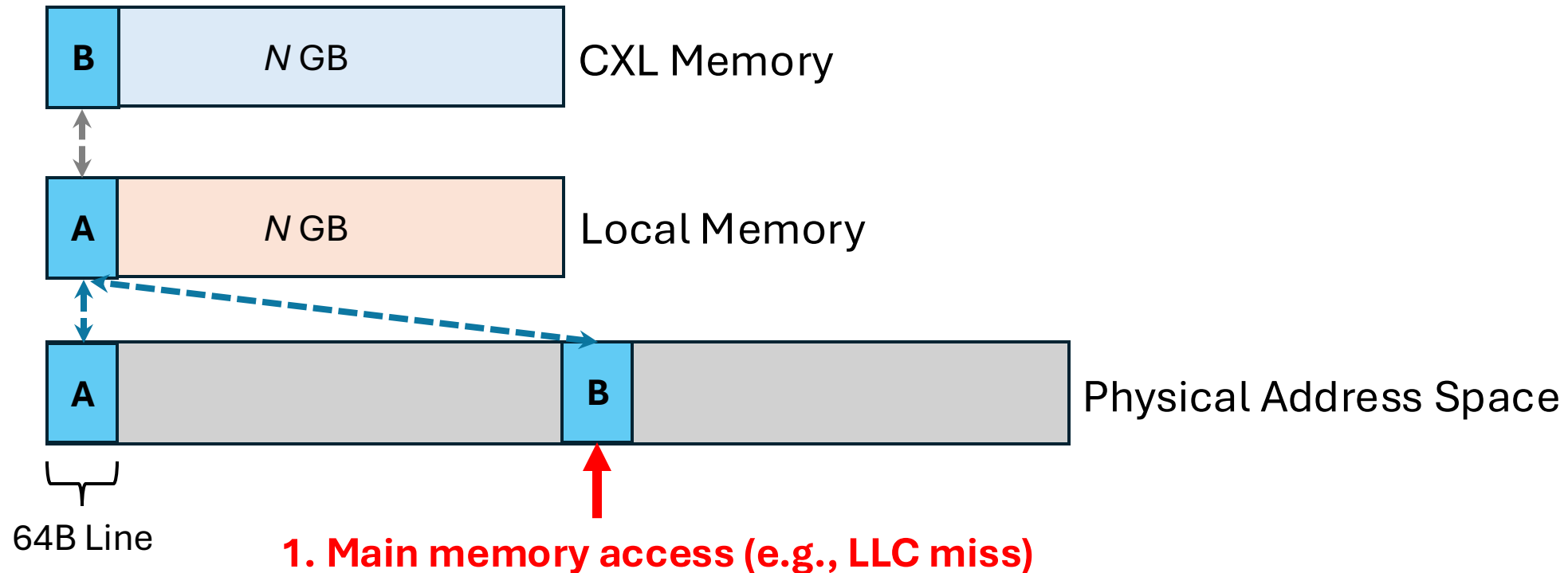
# Associativity and Mapping of Intel Flat Memory Mode

Local memory as a **direct-mapped, exclusive** cache of CXL memory



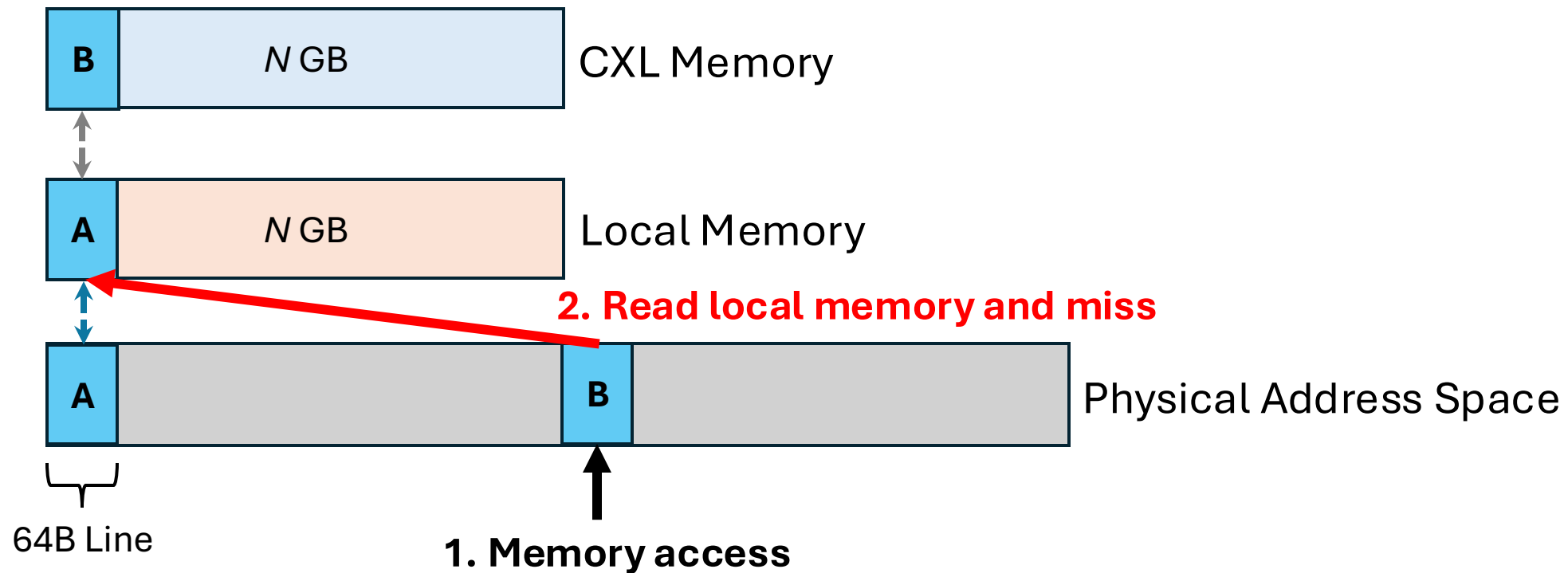
# Local Memory Miss in Intel Flat Memory Mode

When a main memory access misses in local memory, the hardware will “**swap**” the two cache lines



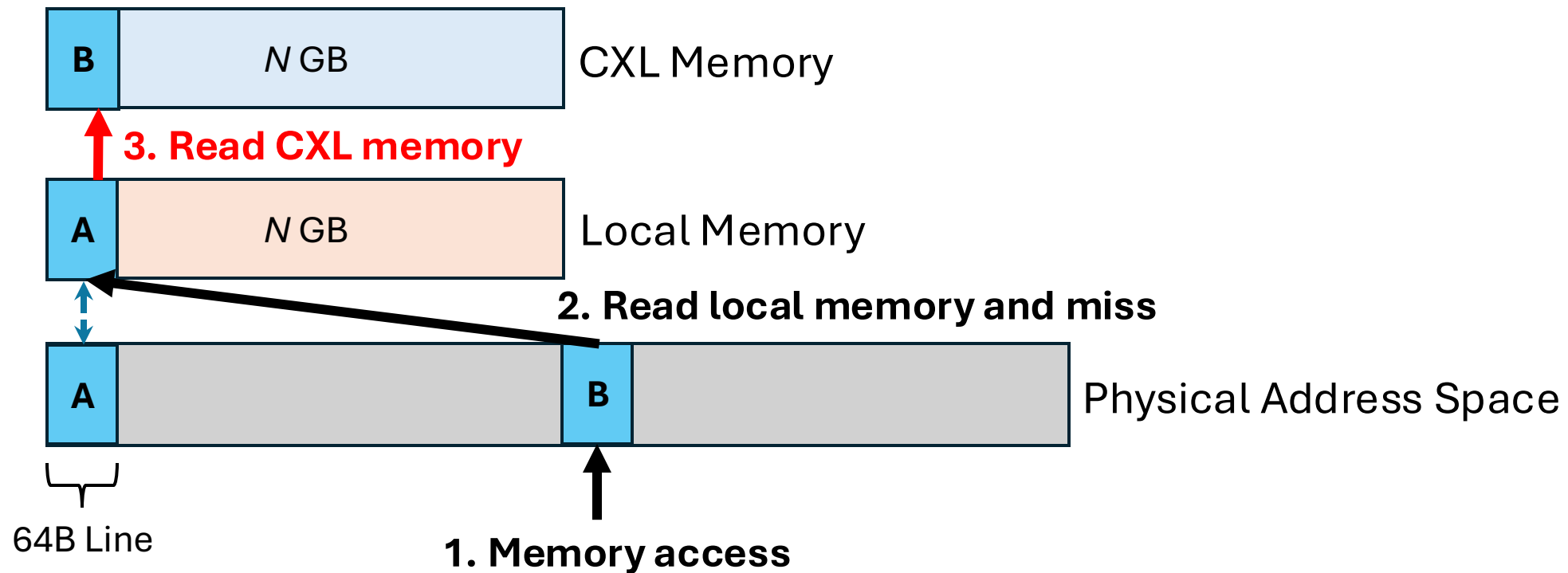
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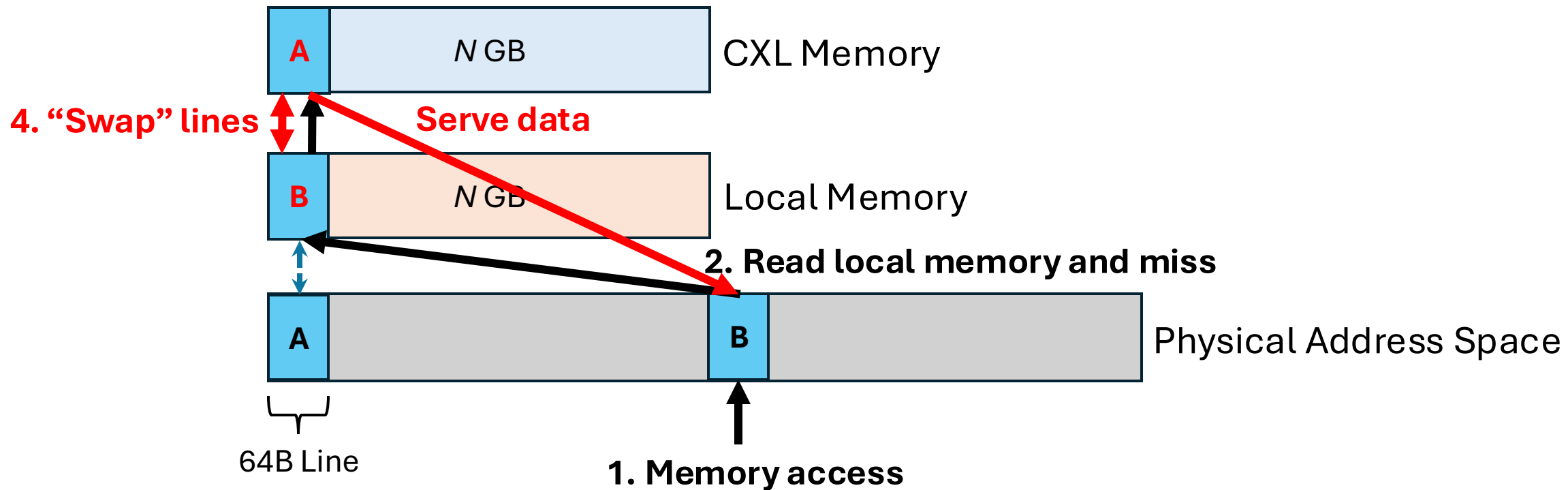
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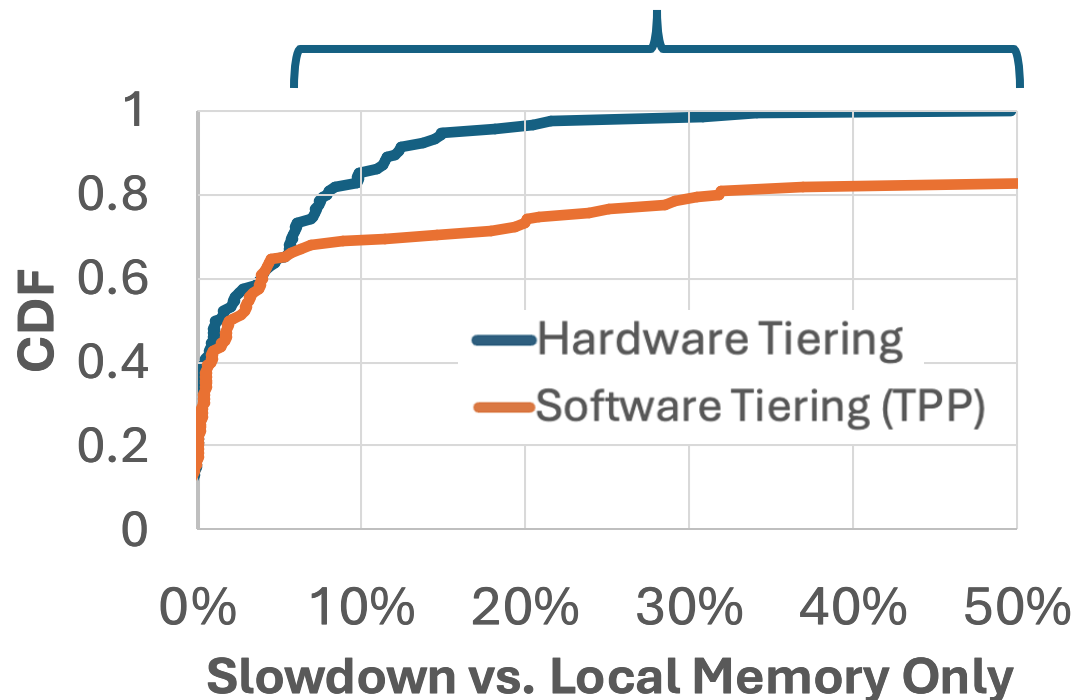
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# Hardware Tiering Alone Still Has Limitations

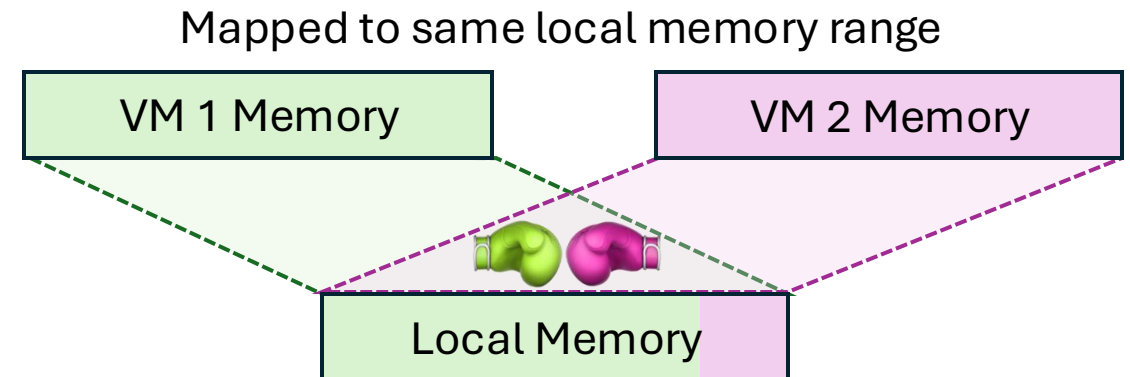
Challenge 1: Some workloads have **heavy local memory misses**

26% workloads have > 5% slowdown  
("outlier" workloads)



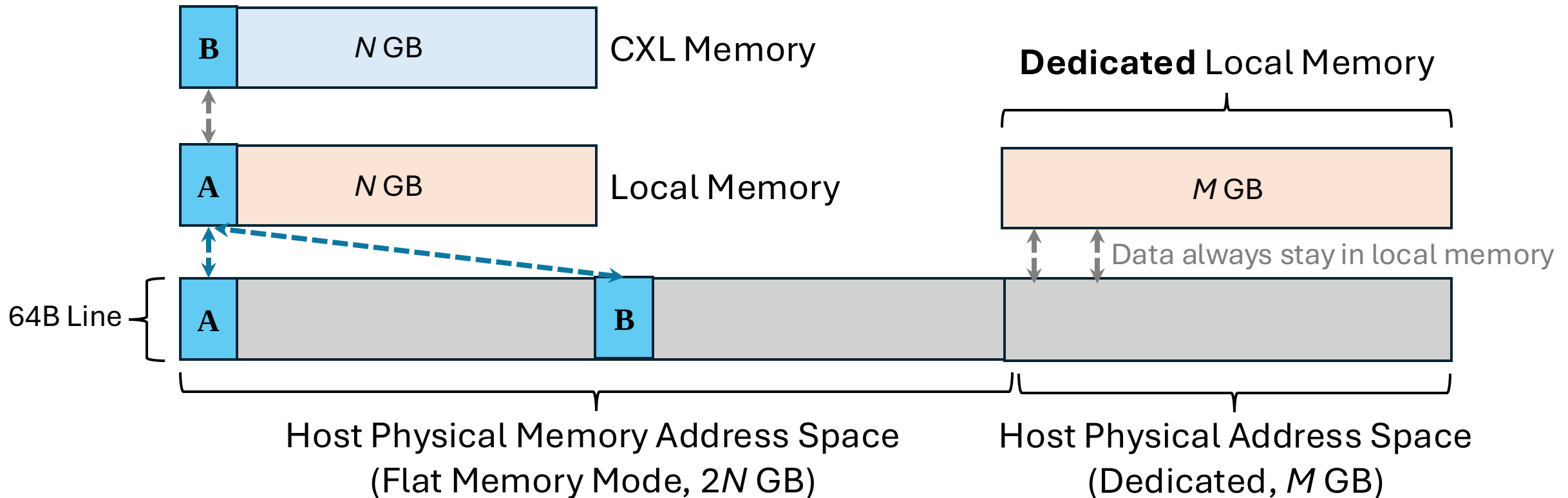
Challenge 2: **No performance isolation** across VMs

**Local memory contention** across VMs  
(more than 50% slowdown)



# Adding Dedicated Local Memory for Outliers

Question: How to allocate dedicated local memory across VMs?



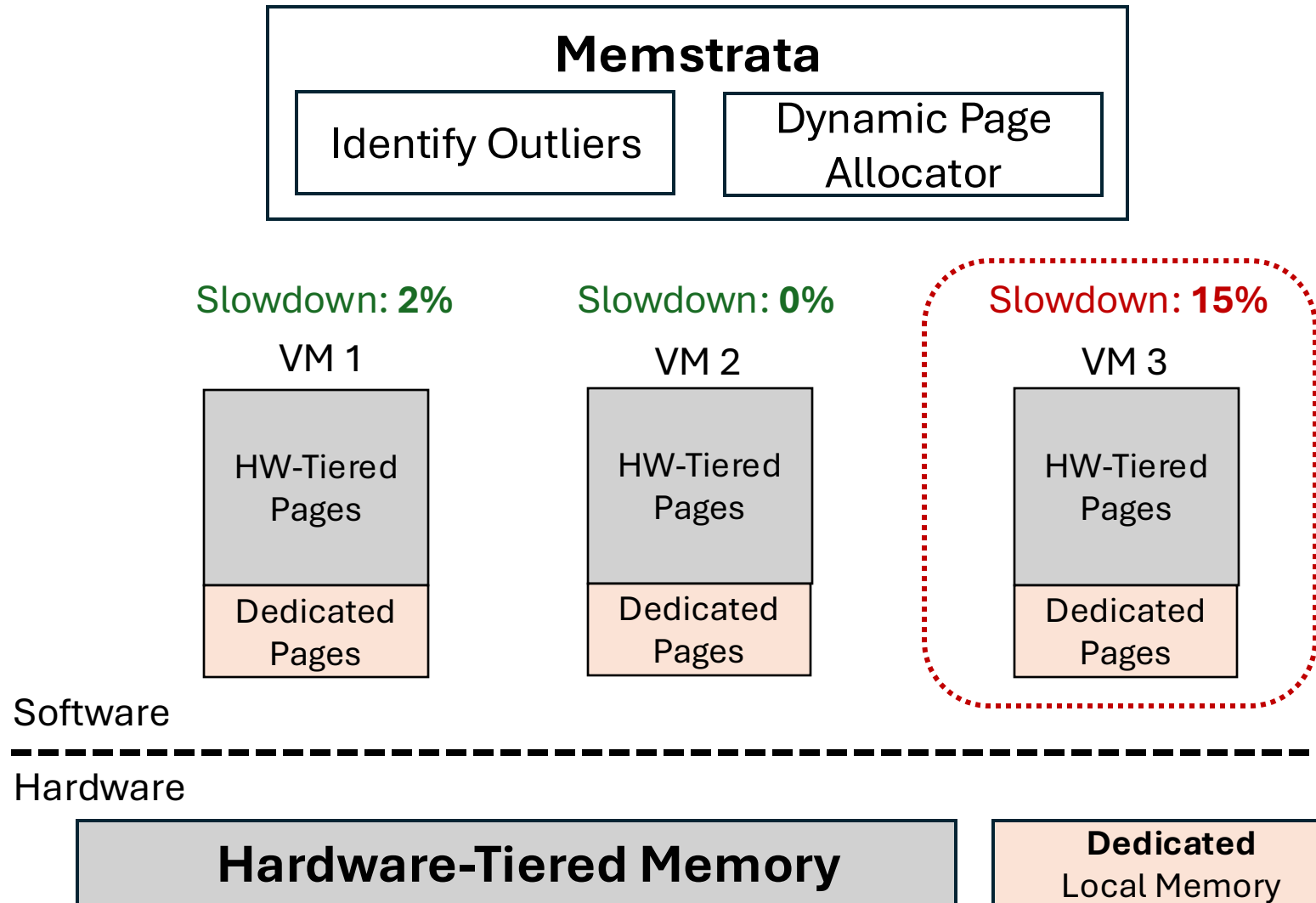


# Memstrata: Memory Allocator for Hardware Tiering

- A lightweight memory allocator in the hypervisor
- Dynamically allocates dedicated memory to **eliminate outliers**
- Provides **performance isolation** between VMs using page coloring

Memstrata + hardware tiering reduces slowdown from 34% to ~5% across all workloads

# Memstrata Dynamically Allocates Dedicated Pages



# Identifying Outliers in Hypervisor Is Challenging

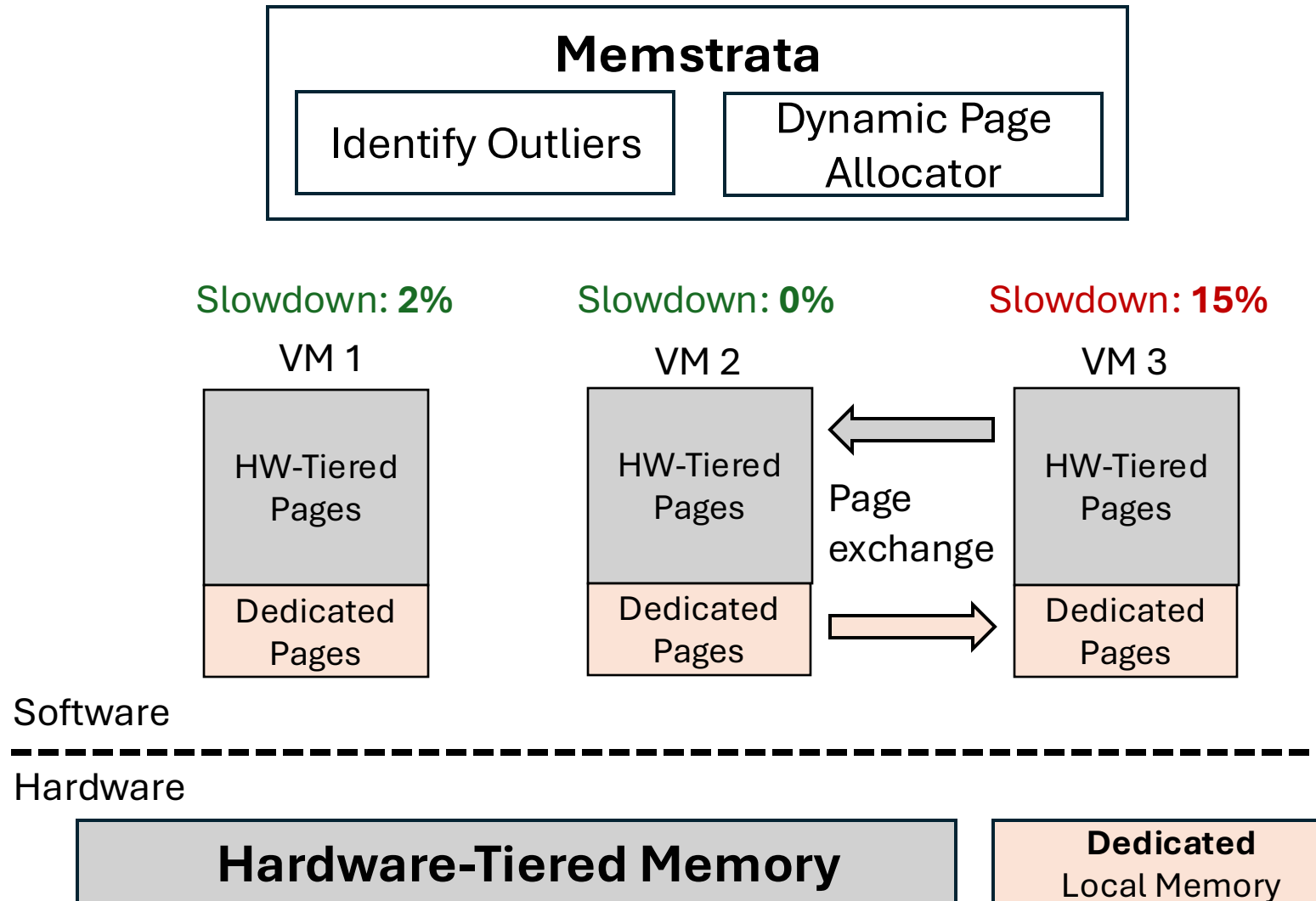
## Challenges:

- Hypervisor is unaware of VM workloads
- Hardware tiering only provides system-wide local memory miss rate

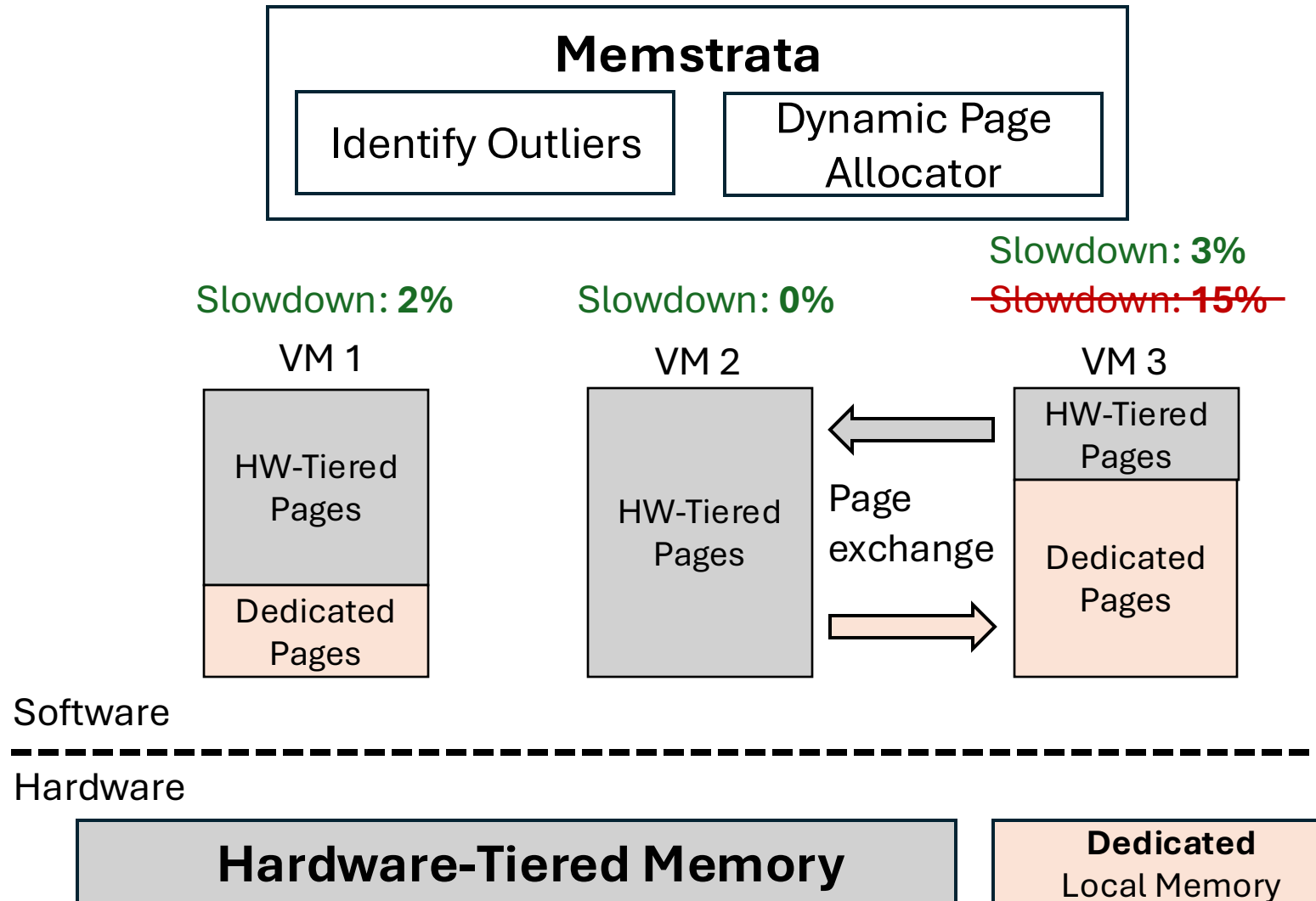
We build a **lightweight prediction model** to identify outliers using low-level performance metrics

- **Per-core** metric: L3 miss latency correlates with miss ratio

# Memstrata Dynamically Allocates Dedicated Pages

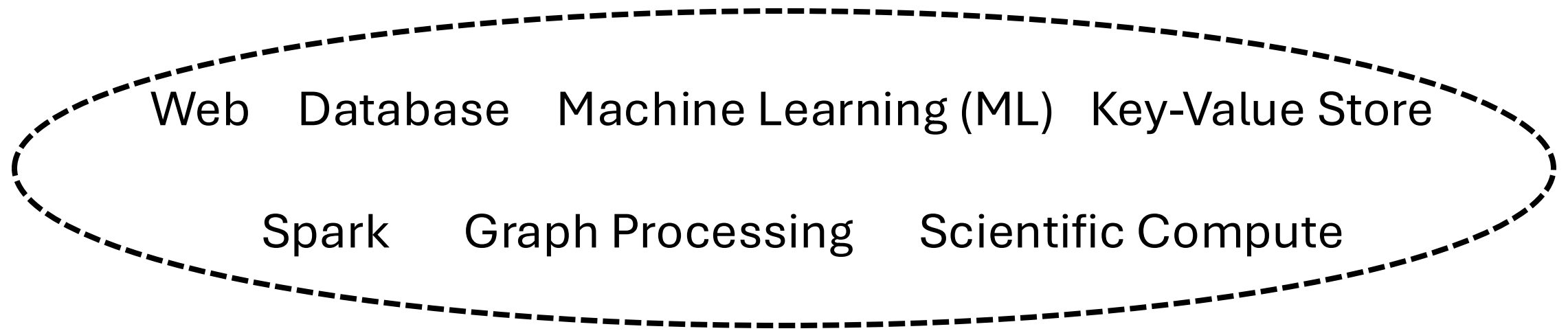


# Memstrata Dynamically Allocates Dedicated Pages



# Evaluate 115 Popular Cloud Workloads

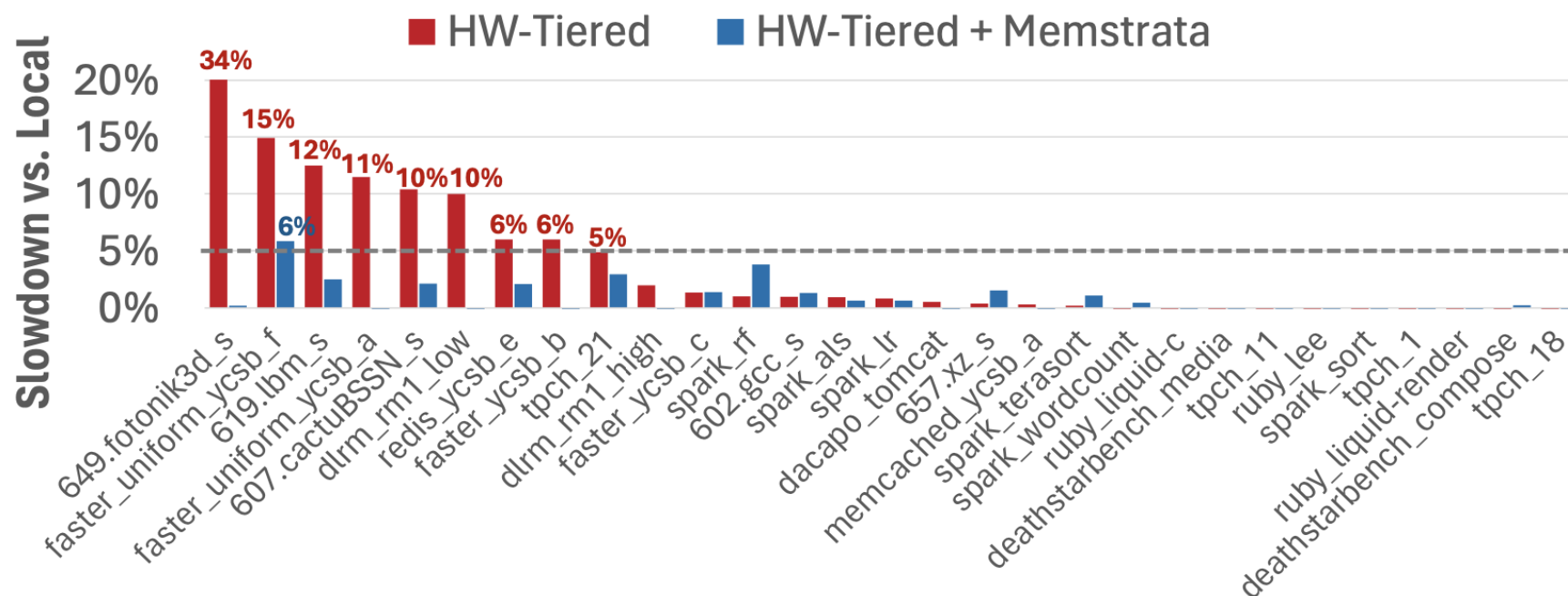
Pre-production Intel Xeon 6 CPU with **real CXL cards** from Astera Labs



115 workloads in total

# Memstrata Eliminates Outliers With Low CPU Overhead

- Sample workloads from representative Azure workload compositions
- Continuous VM arrivals and departures
- Memstrata **mitigates outliers** with **low CPU overhead** (< 3% of a core)



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Source Code

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